Using Natural Language Processing in a job interview to judge employer's required skills from employees

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Abstract*—This paper offers some additional guidelines for MCAST IICT 2nd year B.Sc. students on research paper writing. The abstract is the first paragraph that any researcher reads and thus you need to capture the essence of your research. Dedicate 1 sentence for the theme (subject matter), another for your project aim, one for the proposed solution, then for the general outcome of this project (positive/negative result/outcome), another for the technology used (Windows/Linux, Android/iOS, C#/Java, Unity/Unreal), finally for any technique used (Neural Networks, Pearl Noise, Market Basket Analysis, K-Means, HMM, Augmented Reality). Following is an example: *In this research we are tackling the automatic annotation of cast members and use of key props within movies. Movies tend to gather a huge fan gathering who dedicate a lot of time and resources for documenting the content, thus tools to automate processes such as the timeline when cast members appear, presents of props such as weapons and other forms of annotations are desired. In this research we propose the use of image processing techniques, namely Principal Component Analysis (PCA) and Convolutional Neural Networks (CNN) for the automatic annotation of actors' timeline and presence of weapons in the popular Tv series Game of Thrones (GoT). Our proposed solution managed to detect the targets in 85% of the cases and we identify situations where this is challenging and recommend future research directions.

***Index Terms*—Personality, traits, zero shot classifer, scoring**

I. Introduction

The introduction is intended to give a more detailed overview of the research from the abstract. Document the **theme**, the **aim** and **motivation**. By motivation we mean the justification to why this project/research is relevant/needed for the current year/period and for your specific area of studies. You can consider also including your **hypothesis** and **research questions** in this section. Conclude this section with an overview of what to expect in the next sections. It is good practice to cross-reference your own paper using the ref command for the matching section label, such as in Section II a review of current research is found.

Due to shift towards "Industry 4.0", companies were forced to seek automated solutions for hiring. With rise of Transformer-based models using Natural Language Inference (NLI), it allows machines to understand the intent behind a candidate's story. The aim of this research was to develop and validate an interview prototype utilizing Zero-Shot NLP to provide real-time scoring of candidates personality traits.

II. Literature Review

Give a **brief** overview of the theme, history and facts. You should use the present tense since what you are documenting is current knowledge. Do not dedicate a lot to historical facts about the subject matter, but limit to a paragraph. Make sure you quantify by quoting statistics where relevant. Following is an example for a research in "Automatic cyber-bullying detection": *With the popularity of social media, predominantly amongst the adolescent and young adult generations, there appears to be a correlation between the increase in suicide rates and social networking adoption [?], with suicide attempts increasing by 10% amongst adolescents and by 5.6% amongst young adults for the period 2013-2017.*

A. Pansare, P. Panwar, and P. Kosamkar [1] conducted a comparative study on natural language processing systems based on predicting personalities. Researchers used an MBTI dataset containing 8670 rows (personality set and fifty recent tweets) and two columns where it represents mostly all personality types. All tweets collected and questionnaire was used to judge the participants personality. The system, developed by researchers, consists of Logistic Regression, Random Forest Algorithm, Support Vector Machines, and XGBoost. Results found that the system scored high accuracy scores with thinking/feeling attribute being the highest, but lowest being perceiving vs judging.

S. R. Karra, S. T Nguyen, and T. Tulabandhula [2] conducted a comparative study from large language models on judging participants personality based on answers with use of Big Five Factors (Extraversion, Neuroticism, Agreeableness, Conscientiousness, Openness). Researchers used labels (each label has two ends such as from extroversion to introversion) on zero shot classifiers (ZSC), which are large language models, to judge the participants answers. Researchers used pre-processed datasets, from BookCorpus and Wikipedia, containing both sentences and small paragraphs for the ZSC to judge on.

S. Sikström, I. Valavičiūtė, and P. Kajonius [3] analysed the validity of NLP judging personality based on word-based responses. 663 participants were recruited from United States of America, United Kingdom, and other countries with use of Prolific.io which is an online recruitment tool to collect data. The session consists of two phases. Phase 1 was where participants wrote a description about own or someone else's specific personality trait with use of keywords and no mention of personality trait. Phase 2 was where other participants read participant's written description. IPIP-NEO-30, derived from International Personality Item Pool, was used to measure, measured by participants, five personality factors, extraversion, openness, agreeableness, conscientiousness, and neuroticism. Demographic information such as age, gender, birth of country, education and confidence level were collected with use of questionnaires. The user-made descriptions, written in Microsoft Word, were analysed with use of GPT-4 and BERT based on the big five personality traits. Results found higher prediction accuracy than the rating scale, higher reliability, and better observation than self-report.

J. Kurek et al. [4] conducted comparative study on zero shot classifier as a recruitment tool. Dataset contained unique job candidates, unique jobs, and matches between jobs and candidates totalling up to 3169 records. Only English documents were used only used due to researchers focus on English documents only. Models such as all-MiniLM-L2-v2 and LaBSe were selected by the researchers due to "models' ability to produce embedding vector representations that accurately capture semantic meanings". Results found high accuracy scores, importance of selecting the appropriate model as a recruitment tool, high performance, and recommended modifications on pre-trained models.

L. Hu et al. [5] made use of large language model (LLM) to analyse personality based on user's input. Researchers collected user's posts (from Kaggle and Pandora) which was used on LLM, GPT 3.5 Turbo, to analyse from three perspectives (semantic, sentiment, and linguistic perspective). Prompts were made to LLM to generate explanation on each MBTI trait and calculate similarity between the user and the label description with use of soft labels. Results found that the model, constructed by the researchers, outperformed all traditional methods that detect personalities, performance impact from focus on linguistic over semantic, and struggles to generate explanations with use of LLMs rather than specific small models. Limitations include limited dataset, and input length.

All sources touched on use of LLMs and NLPs in the context of job recruitment. Referenced studies made use of GPT-4, BERT, and BART as zero shot classifiers, and Logistic Regression, Random Forest, and XGBoost to predict personality traits from social media datasets. Pansare et al. [1] made use of specific models such as XGBoost, and Random Forest resulted in high accuracy but required large and pre-labelled datasets. On the other hand, Karra et al. [2] and Kurek et al. [4] employed Zero-shot classifiers showed that personality scoring was able to be executed without specific training, supported by Sikstrom et al. [3] who found that GPT-4 offered high reliability than self-report scales. Although LLMs outperform traditional methods, LLMs often struggle to generate explanation based on traits compared to smaller specific models. In addition, Kurek et al. [4] identified a linguistic gap due to limiting found documents to only English, as well as Hu et al. finding performance gaps when focus shifts to linguistic perspectives over semantics. Limitations found in this literature review represent a knowledge gap that the research gap needed to be addressed in this research by using BART-MNLI to evaluate and score open-ended candidate stories.

III. Research Methodology

This section is one of the most important parts of your research and it is the one that will receive most criticism. The reason is that here you document how you conducted your research. Since you are reading a B.Sc. degree you are expected to follow a scientific method, thus in this section you are expected to document: **the problem**; **hypothesis**; **aim & objectives**; **research questions**; and **research pipeline**. Let's go through each.

First start by providing a brief overview of the problem, similar to what you already mentioned in the previous sections. Then provide a hypothesis, this is the assumption you are making, which via your research you are attempting to prove or disprove. An example hypothesis in traffic management would be: *There is a significant positive correlation between the semantic scores generated by the BART-MNLI model and the evaluation scores provided by expert human recruiters for the same set of open-ended responses.*

After explaining the problem and hypothesis you need to specify why you are doing this research and what you think the output will be, so the aim & objectives. Let us consider the "Marine Vessel Tracking" concept: *With this research we aim to be able to automatically detect any anomalies within Marine traffic, such as drug smuggling, human trafficking and/or collisions.* Our objectives are:

1) Collect and simulate data set of open ended job interview responses related to specified employer skills
2) Implement BART-MNLI to generate a score for each user response
3) Conduct correlation analysis between AI scores and human scores
4) Measure and compare time taken to process automated scoring against time taken by human recruiters to evaluate users answers
5) Conduct error analysis on instances with low correlation to detect linguistic issues or terms where ZSC fail to score.

So lastly you have to present a number of research questions that you attempt to answer, which ideally revolve around the **data set**, **algorithm** and **evaluation** which you documented in your literature review:

1) To what extent does the BART-MNLI zero-shot classification model correlate with expert human recruiter eval-

uations when scoring specific employer-required skills in open-ended interview responses?
2) Does the use of automated NLI-based scoring significantly reduce the time required for the initial candidate screening phase compared to traditional human-led qualitative analysis?
3) What are the primary linguistic limitations of the BART-MNLI model when interpreting nuanced or industry-specific terminology in candidate "stories"?

Finally, you need a plan on how you intend to answer the research questions which will help you prove/disprove the hypothesis. Different research areas would require specific pipelines, yet a generic one is shown in Fig 1. After presenting an illustration of your pipeline it is recommended to documenting what has been done at every stage. Remember you should use past tense in this section since you are document what you have already done, since this paper will be published at the end of your research.

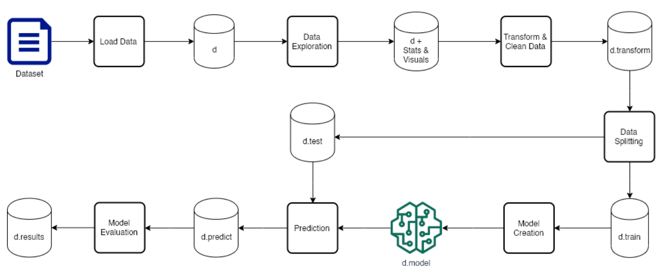

Fig. 1. Research Pipeline

IV. Findings & Discussion of Results

This section is a delicate section and you are encouraged to be as honest and open as possible. The aim is not to show that you got a perfect solution to a long existing problem. Trying to state that in a few months you developed a perfect solution with 100% is not convincing and raises doubts. It is recommended that you document your findings in every step of your research pipeline, highlighting your observations and decisions taken. Present your results and very importantly compare with existing research which you documented in Section II.

The focus of the Project module is for you to delve into an area that exposes you to new technologies and offers you an opportunity to be critical of your work. So you are expected to document where your solution/research worked and where it did not. Reflect and document reasons why the solution/research did not perform as expected and propose ways of addressing this. From these observations you will produce new research questions in the next section. Consider the research in "Brand usage detection within audio streams", where certain key terms were searched within videos, the results of which are documented in Table I. You will notice that the terms "Peppa" and "Sushi" were the least recognised terms even by the best transcribers. Upon investigation we determined that "Peppa" was not recognised cause of voice morphing to create childish voices in the cartoon video, whilst "Sushi" was pronounced by a Japanese person speaking English. So the research student decided to focus his dissertation research on how to create a system that is able to recognise heavy accents to automate the configuration of a transcriber, in this case to cater for English spoken by a Japanese person, which accent is very different from an Indian accent, British accent or Italian accent, just to name a few.

TABLE I
Recall Results

#	Term	Google Cloud	Google Speech	Sphinx CMU
1	Peppa	27%	33%	0%
2	Peppa	33%	22%	0%
3	Apple	96%	92%	79%
4	Galaxy	100%	100%	100%
5	Galaxy	95%	95%	80%
6	Sushi	75%	35%	0%
	Average	71%	62%	43%

V. Conclusion

So the conclusion is most probably the second section that a reader would use to consider reading in full your research. Thus it is important to highlight the essence of your research. The recommended approach is to answer your research methodology. Start by answering your research questions, then stating to what degree did this research achieve its aim and objectives, highlighting potential causes for not being able to do so at a desired level, such as time, or other circumstances. Consider the following: *A student was due to research the use of MCAST computers during out-of-office hours to offer a private cloud computing service for research, similar to Google Cloud but free for MCAST students. Due to the lockdown by COVID-19 pandemic we could not continue on the original planned research objective and had to adapt.*

The final and most important part of the conclusion are your recommendations for future research, not necessarily for yourself (referring to what you plan to do in your dissertation or beyond), but also to other future researchers who might consider doing similar work to yours. The recommendations you provide here will set such prospective researchers on a better track/direction thanks to your experience.

Appendix A
Supporting Material

You can add screen shots and statistics. Stick to essential information.

Acknowledgement

You can dedicate this section for special assistance that you were given by a 3rd party. Not your mentor, relatives or questionnaire participants.

REFERENCES

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